# Validity Is Not Difficulty: Hazard-Preserving Physical 3D World Compilation for Autonomous Driving

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## Abstract

*Driving-world reconstruction systems often conflate two causes of difficulty: a scene may be hard because it contains a legitimate hazardous actor, or invalid because reconstructed geometry contradicts sensor evidence. Treating both as low confidence can obtain superficially safer worlds by deleting precisely the actors that matter. We introduce HARP-3D, a hazard-preserving physical 3D world compiler that separates appearance from collision geometry and artifact validity from behavioral hazard. The compiler aligns multi-frame Actor evidence in canonical coordinates and exposes four surface actions—*KEEP*,* PROJECT*,* COMPLETE*, and *UNKNOWN*—without changing Actor identity, trajectory, or extent. On frozen Argoverse 2, ray-certified compilation eliminates paired free-space violations, reduces target LiDAR depth error from 0.207 to 0.154 m and symmetric Chamfer from 0.254 to 0.175 m, and retains all Actor and hazard state. A nearest-surface alternative reaches 0.168 m Chamfer but leaves 84.9% matched-ray violations, exposing a geometry–safety boundary. A nuScenes-only factorized repair-or-abstain head then covers 75.6% of AV2 Actors zero-shot, reduces false repair from 16.6% under always-repair to 9.0%, and obtains 0.182 m selective Chamfer with 0.981 hazard AUROC and zero cross-input shift. On exact-match source Actors it rejects every geometrically harmful repair, yet its selected Actors have 2.28× larger long-tail motion error, showing why repair and trajectory authority must remain separate. Post-hoc attribution further assigns 49.5% of AV2 validity sensitivity to observation-window length. We therefore report empirical transfer while explicitly withholding a domain-invariant, conformal, or road-safety guarantee.*


## 1. Introduction

Reconstructed driving worlds are increasingly used for perception analysis, counterfactual replay, and planning evaluation. Their geometry, however, is rarely a literal collision model. A rendered Gaussian may explain image appearance while occupying free space, duplicating an Actor shell, or flickering across time. Conversely, a perfectly valid Actor may be difficult because it cuts in, brakes hard, or crosses the Ego path. A useful world compiler must distinguish these cases: *world validity is not world difficulty*.

This distinction matters because a generic confidence filter has an easy but scientifically undesirable failure mode. It can improve aggregate collision proxies by suppressing uncertain or nearby Actors. Such a world looks safer because legitimate hazards disappeared, not because its physical state became more faithful. We instead ask whether evidence-inconsistent geometry can be repaired while identity, lifecycle, trajectory, size, time-to-collision, and hazard events remain fixed.

We propose HARP-3D, a physical compilation layer between a reconstructed scene and downstream task queries. It represents a world as

$$\mathcal{W}_t = (\mathcal{G}_t^{\mathrm{app}}, \mathcal{S}_t^{\mathrm{phys}}, \mathcal{A}_t, \mathcal{P}_t, \mathbf{q}_t^{\mathrm{val}}), \quad (1)$$

where appearance Gaussians and physical collision surfaces have separate semantics. The compiler uses multi-frame Actor alignment, ray free/occupied evidence, temporal support, and source provenance to select one of four local actions. Unsupported regions become UNKNOWN, never free by default, while Actors remain in the scene.

Our second component factorizes validity from hazard with disjoint inputs and paired counterfactual diagnostics. Changing a clean Actor from ordinary driving to a legal near-miss should not change its repair score; injecting a duplicate shell or temporal pop should not change its hazard score. A selective interface repairs an Actor only when nuScenes-calibrated runtime evidence supports it and otherwise returns the clean query. Our third component links physical state quality to downstream reliability. An Actor residual distribution is projected onto a candidate trajectory boundary, yielding a continuous cost whose conditional density supports arbitrary budgets and multiple horizons.

Our contributions are:

- a hazard-preserving physical compiler with explicit KEEP/PROJECT/COMPLETE/UNKNOWN semantics and separate appearance/collision state;
- a nuScenes-only repair-or-abstain factorization protocol that measures cross-input leakage and reports its AV2 risk–coverage boundary;
- a continuous task-cost density and monotone multi-horizon reliability interface, with an exact-identity audit that keeps physical-repair and trajectory authority distinct.

## 2. Related Work

**Driving-world reconstruction.** Multi-sensor driving datasets such as nuScenes [1] and Argoverse 2 [8] provide calibrated imagery, LiDAR, ego pose, maps, and tracked Actors. Neural scene graphs model dynamic objects explicitly [4], while 3D Gaussian Splatting provides a high-quality appearance representation [2]. Our work does not reinterpret Gaussian opacity as collision occupancy. It adds a separate, queryable physical surface with sensor provenance.

**Uncertainty and reliability.** Deep ensembles expose useful empirical predictive disagreement [3], and conformalized quantile methods can turn residuals into calibrated marginal intervals under appropriate assumptions [5]. Driving logs violate many simple exchangeability assumptions through scene correlation and domain shift. We therefore report empirical calibration explicitly, preserve claim boundaries, and use monotone structure only for runtime consistency—not as a formal road-safety guarantee.

**Our distinction.** Prior confidence, occupancy, and reconstruction-quality signals often mix physical invalidity, visibility, and behavioral difficulty. HARP-3D instead defines separate evidence inputs, paired interventions, and failure denominators so that an improvement cannot be obtained by deleting hazardous Actors or relabeling unknown space as free.

## 3. Problem Formulation

Let Actor $i$ have an immutable semantic state

$$A_i = (ID_i, X_{i,0:H}, D_i), \tag{2}$$

containing identity, trajectory, and dimensions. Appearance primitives may be associated with $A_i$, but collision queries consume only an Actor-local physical surface $S_i^{\text{phys}}$ and its known/unknown mask.

We separate two random variables. Artifactness $a_i$ depends on sensor and geometric evidence $E_i$, physical surface state, and provenance:

$$a_i = P(\text{artifact} \mid E_i, S_i^{\text{phys}}, P_i). \tag{3}$$

Hazardness $h_i$ depends on Actor dynamics, map context, and a candidate Ego trajectory $\tau$:

$$h_i = P(\text{hazard} \mid X_{i,0:H}, \tau, \mathcal{M}). \tag{4}$$

The goal is paired conditional invariance, not unconditional statistical independence. For the same clean Actor, a legal safe-to-hazard intervention should preserve $a_i$. For the same behavior, a clean-to-artifact intervention should preserve $h_i$.

For task reliability, let $n_{\tau,t}$ be the local boundary normal, $d_{i,t}(\tau)$ the signed Actor clearance, and $\epsilon_{i,t}$ the Actor residual. We define

$$C(\tau, H) = \max_{i,t \le H} \frac{|n_{\tau,t}^{\top} \epsilon_{i,t}|}{\max(|d_{i,t}(\tau)|, \varepsilon)}. \tag{5}$$

The runtime interface returns $R(\tau, H, b) = P(C \le b)$ and enforces $\partial R / \partial b \ge 0$ and $\partial R / \partial H \le 0$.

## 4. Method

### 4.1. Actor-Preserving Physical Compilation

For each Actor, we transform LiDAR endpoints, rays, and associated primitives into a canonical Actor frame,

$$x_{i,t}^{\text{actor}} = T_{\text{actor},t}^{-1} T_{\text{ego},t} T_{\text{sensor}} x_t^{\text{sensor}}. \tag{6}$$

The fused representation stores surface support, observed-free intervals, temporal counts, local normals, viewing-direction diversity, and provenance. Compilation is deterministic and local:

- KEEP: stable sensor-supported surface;
- PROJECT: a near-surface primitive contradicts observed free space and is projected to supported geometry;
- COMPLETE: a local hole has sufficient multi-frame and multi-view support;
- UNKNOWN: evidence is missing, contradictory, or insufficient.

Unknown is a first-class state and never silently becomes free. All four actions preserve $(ID_i, X_{i,0:H}, D_i)$. Appearance primitives may attach to the repaired surface using center distance, covariance-normal alignment, depth, and visibility, but physical queries do not read Gaussian opacity.

### 4.2. Validity–Hazard Factorization

The validity encoder receives ray contradiction, surface residual, temporal support, provenance, visibility, and local geometry. Its label is whether the frozen compiler's target Chamfer is no worse than that of the untouched clean query. The hazard encoder receives only TTC, clearance, closing speed, braking, and crossing geometry. Our factorized model uses two structurally disjoint low-capacity heads; a shared-input two-head network is the learned baseline:

$$\mathcal{L} = \text{BCE}(s_{\text{rep}}, y_{\text{rep}}) + \text{BCE}(s_{\text{haz}}, y_{\text{haz}}). \tag{7}$$

Multi-frame Actor alignment ray FREE/OCC evidence → Physical compiler KEEP / PROJECT COMPLETE / UNKNOWN → Validity–hazard paired factorization → Actor residuals analytic task query → Cost density monotone reliability

Figure 1. HARP-3D compiles sensor evidence into an Actor-preserving physical world, separates validity from hazard, and exposes task-conditioned reliability. Appearance Gaussians never serve directly as collision occupancy.

We fit features and weights on nuScenes train only. On a disjoint calibration fold, threshold $q$ is the maximum-coverage value satisfying the population false-repair bound

$$\widehat{R}^{+}(q) = \frac{1 + \sum_{i=1}^{n} \mathbf{1}[s_i \geq q,\, y_i = 0]}{n+1} \leq \alpha. \tag{8}$$

Accepted Actors use the compiled surface; abstained Actors retain the clean query. Paired input swaps measure whether either score reads the other task's variables. The finite-sample interpretation requires exchangeability; AV2 uses the frozen $q$ only for empirical zero-shot evaluation.

### 4.3. Actor Residuals and Boundary-Cost Density

For Actor $i$ and horizon $t$, a low-capacity head predicts a 2D residual distribution $(\mu_{i,t}, \Sigma_{i,t})$. Analytic projection onto $n_{\tau,t}$ gives mean $n^\top \mu$ and variance $n^\top \Sigma n$. Equation 5 converts these residuals and deterministic clearance into a continuous trajectory cost.

We model $y = \log(1+C)$ with a small conditional Gaussian mixture,

$$p_\theta(y \mid z^{\text{val}}, z^{\text{haz}}, \tau, H) = \sum_k \pi_k \mathcal{N}(y; \mu_k, \sigma_k^2), \tag{9}$$

which exposes a closed-form marginal CDF for any budget. A low-rank, input-conditioned dependence model combines the frozen marginals across horizons. This factorization separates learned Actor uncertainty from analytic task geometry.

### 4.4. Monotone Runtime Surface and SceneIR

The final runtime object is a frozen grid over horizon and cost budget with monotone interpolation. Budget CDF values are non-decreasing; prefix-horizon reliability is non-increasing. Groupwise isotonic maps may calibrate predefined conditions on a disjoint source fold, but do not change these structural constraints.

The two authorities compose without implication. The repair gate first chooses the surface representation,

$$\widetilde{S}_i = g_i S_i^{\text{comp}} + (1 - g_i) S_i^{\text{query}}, \qquad g_i = \mathbf{1}[s_i^{\text{rep}} \geq q]. \tag{10}$$

The downstream task separately grants authority only if its multi-horizon reliability satisfies the requested budget, $a_{i,t}(c) = \mathbf{1}[R_i(t, c \mid \widetilde{S}_i, \tau) \geq \rho]$. In particular, $g_i = 1 \not\Rightarrow a_{i,t}(c) = 1$: evidence can support a physical repair while Actor motion remains uncertain.

SceneIR stores Actor identities and transforms, physical surfaces, known/unknown masks, provenance, and validity fields. The AV2 adapter performs only coordinate, unit, timestamp, category, and sensor-opportunity normalization. It does not fine-tune, recalibrate, or change thresholds using AV2 quality.

## 5. Experiments

**Data and protocol.** We train and select models only on nuScenes. Argoverse 2 Sensor validation logs form a zero-shot external test. Before reading any method output, we sort the 150 validation UUIDs and select every fifth log, yielding 20 quantitative and 10 qualitative logs. No AV2 fine-tuning, post-hoc calibration, threshold selection, or failed-scene deletion is allowed. Final nuScenes and AV2 cohorts are evaluated exactly once after the method freezes.

**Baselines.** For physical compilation we compare naive reconstruction, a legacy validity filter, V6.7 inward-ray repair, and HARP-3D. For factorization we compare a shared encoder, a two-head shared trunk, independent encoders, and paired factorization. Reliability baselines include a scalar confidence, independent horizon classifiers, a continuous marginal density, and the full joint multi-horizon surface.

**Metrics.** Physical metrics include free-space violation, ray termination, point-to-surface error, LiDAR depth consistency, temporal jitter, ghost components, and surface completeness. Hazard preservation reports Actor/ID/lifecycle retention, trajectory and TTC shift, and event-count change. Factorization reports artifact/hazard AUROC and AUPRC, clean-hazard false-artifact rate, paired prediction shift, and cross-probe leakage. Reliability reports log likelihood, integrated Brier score, calibration error, Spearman correlation, monotonic violations, and runtime.

**Zero-shot Actor-local surface evidence.** Following the dynamic-Actor canonicalization boundary used by NeuRAD [7], we transform ego-frame AV2 LiDAR points inside each annotated cuboid into its Actor frame. Unlike appearance-oriented seed initialization, we never mirror points: every collision surfel must retain sensor and temporal support. We fuse two of every three frames at 0.12 m


Figure 2. The first metadata-frozen AV2 main case. From left: synchronized RGB with observed Actor returns, paired synthetic artifacts, the four-action ray-certified output, and motion-compensated sparse camera depth. Camera/crop selection does not read RGB appearance or method quality. This is a calibrated evidence overlay, not photorealistic reconstruction.

Table 1. nuScenes-only selective repair and AV2 zero-shot transfer. False repair is measured over all Actors; coverage is the accepted fraction. Chamfer is for the repair-or-clean-query output. Leak is the maximum mean prediction shift under a paired swap of the other task's input. Each head's nuScenes-calibrated threshold is reused unchanged on AV2.

| Dataset | Head | Repair AUROC ↑ | Hazard AUROC ↑ | Coverage ↑ | False repair ↓ | Selective CD ↓ | Max leak ↓ |
|---|---|---|---|---|---|---|---|
| nuScenes test | shared | 64.26 | 91.66 | 11.84 | 3.51 | 0.231 | 28.86 |
| nuScenes test | factorized | **64.91** | **98.11** | 3.95 | **0.44** | 0.246 | **0.00** |
| AV2 zero-shot | shared | **77.97** | 92.89 | 55.21 | **3.15** | 0.190 | 23.46 |
| AV2 zero-shot | factorized | 69.96 | **98.05** | **75.55** | 8.99 | **0.182** | **0.00** |

| AV2 Actor surface | Dist. (m) ↓ | Recall (%) ↑ |
|---|---|---|
| Single observed frame | 0.864 | 26.16 |
| Multi-frame canonical surfels | **0.131** | **85.72** |

Table 2. Zero-shot held-out LiDAR support on the frozen 20-log AV2 quantitative cohort (1,046 Actors, including 241 hazardous Actors). The comparison uses real sensor points; paired synthetic corruptions are evaluated separately as a compiler contract.

| Surface | Free (%) ↓ | Recall (%) ↑ | CD (m) ↓ |
|---|---|---|---|
| Single-frame input | 100.0 | 53.36 | 0.254 |
| Nearest canonical | 84.91 | **62.60** | **0.168** |
| Ray-certified | **0.00** | 59.56 | 0.175 |

Table 3. Frozen zero-shot AV2 quantitative result. Nearest-surface projection fits geometry best but violates matched-ray free space; ray-certified projection accepts a small Chamfer cost. Later held-out frames are target-only.

and reserve the remaining frames as unseen target geometry. Across 1,046 Actors and 1.075M stable surfels, fusion reduces the mean target-to-surface distance from 0.864 m to 0.131 m and raises recall at 0.20 m from 26.16% to 85.72% (Table 2).

The paired corruption atlas contains clean, observed-free ghost, duplicate-shell, temporal-flicker, and supported-hole probes without changing Actor identity, trajectory, size, or hazard. It yields 100.0% artifact recall, 0.00% clean-hazard false-artifact rate, and 100.0% Actor/hazard retention. These perfect paired scores verify the deterministic compiler contract and input separation; they are not reported as natural-artifact generalization.

**Four-action physical compilation.** We next freeze one single-frame query per Actor and reserve all later held-out frames as target-only geometry. The compiler reads only the build surface and query evidence. A nearest-canonical projection initially reduces Chamfer to 0.168 m, but ray-specific evaluation reveals 84.91% early terminations in the paired observed-free interval. Our final operator is ray-certified: a primitive with direct LiDAR provenance projects to its matched observed termination; without one it becomes UNKNOWN. Across 433 quantitative Actors (147 hazardous), this reduces paired free-space violations from 100.0% to 0.00%, target depth error from 0.207 m to 0.154 m, and Chamfer from 0.254 m to 0.175 m, while raising ray-termination consistency from 56.94% to 64.38% (Table 3). Actor identity, trajectory, extent, and hazard labels are retained for 100.0% of Actors. The zero free-space rate is a paired contract result; depth, ray termination, and Chamfer use independent target-only LiDAR.

**Frozen camera evidence and an exposed boundary.** We project each pre-registered qualitative Actor back into syn-

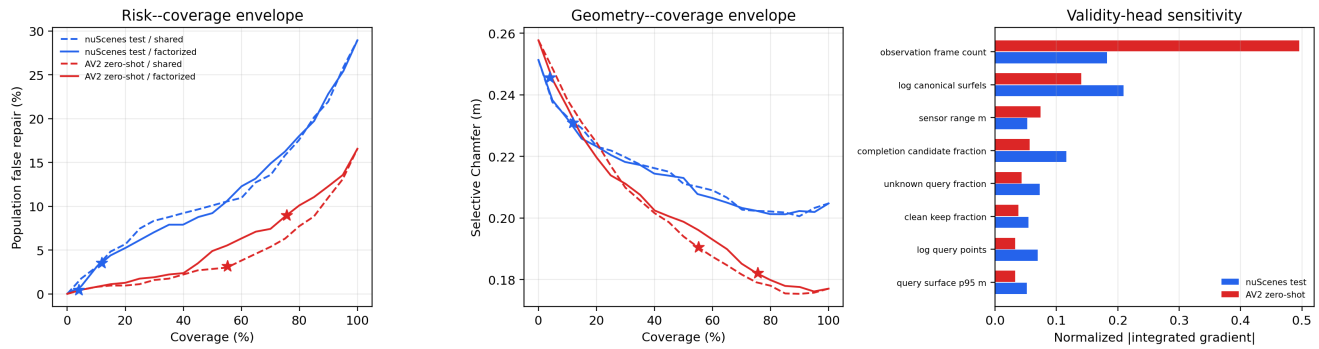


Figure 3. Frozen P7 safety envelope. Left and center: risk–coverage and geometry–coverage curves; stars mark the unchanged nuScenes-calibrated thresholds and never select an AV2 operating point. Right: Integrated Gradients relative to the nuScenes-training mean [6]. AV2 reliance on observation-frame count exposes a sensor-opportunity shortcut. Curves and attributions are descriptive rather than a cross-domain certificate.

chronized AV2 ring-camera frames using the official calibration and motion compensation. Camera identity maximizes only the number of visible query LiDAR returns, with a fixed camera-order tie break; RGB is decoded afterward and every case is retained. All 30 cases have calibrated RGB, sparse depth, four-action, and before/after video evidence; the median/minimum visible query counts are 82/14 (Fig. 2). Importantly, the frozen set exposes a limit: although mean Chamfer improves, 17.19% of all 634 Actors worsen individually. Thus P3 establishes aggregate zero-shot geometry, not a per-Actor repair certificate; this motivates a selective "repair or abstain" validity head trained and calibrated only on nuScenes.

**nuScenes-to-AV2 selective factorization.** The strict multi-frame surface contract yields 29/56/228 Actors for nuScenes train/calibration/test; we retain the small training set rather than loosen admissibility. On nuScenes test, factorization is non-inferior to the shared head for repairability (64.91 vs. 64.26 AUROC) and raises hazard AUROC from 91.66 to 98.11, while maximum paired cross-input shift falls from 28.86% to 0.00%. The conservative threshold covers 3.95% with 0.44% population false repairs.

Without AV2 fitting, the same selector covers 75.55% of 634 Actors and reduces false repair from 16.56% under always-repair to 8.99%. Clean-query/always/selective Chamfer is 0.258/0.177/0.182 m. Thus abstention costs only 5 mm mean Chamfer relative to always-repair while nearly halving false repairs. Hazard AUROC is 98.05, hazard coverage is 92.17%, and factorized cross-input shift remains zero. However, coverage changes from 8.93% on nuScenes calibration to 75.55% on AV2; Table 1 therefore supports empirical transfer, not a cross-domain conformal guarantee.

Table 4. Frozen physical–trajectory interface on exact-match nuScenes test Actors. Cost and error are row-level 90th percentiles over retained V6.7 outcomes; FS/flip gives event counts. Physical selection is not trajectory authority.

| P4 group | Actors | q90 cost ↓ | q90 error ↓ | FS / flip ↓ |
|---|---|---|---|---|
| selected | 5 | 0.283 | 1.577 | 0 / 0 |
| abstained | 83 | 0.102 | 0.693 | 7 / 47 |
| geometric harm | 23 | 0.076 | 0.543 | 0 / 4 |

**Interpretable safety envelope.** We retain the model, standardizer, thresholds, and all Actor rows and evaluate 21 fixed coverage levels without refitting or recalibration. The factorized score shifts from mean/median $.761/.945$ on nuScenes calibration to $.978/.999992$ on AV2 (Wasserstein .217, KS .704), explaining the operating-point coverage jump. Integrated Gradients [6] relative to the nuScenes-training mean attributes 18.25% of absolute validity sensitivity to observation-frame count on nuScenes test but 49.53% on AV2 (Fig. 3). Structural cross-input derivatives remain exactly zero: task factorization prevents hazard leakage, but does not make validity inputs sensor invariant. We therefore preserve the empirical P4 result while rejecting the stronger interpretation of a domain-invariant selector.

**Physical repair is not trajectory authority.** We exact-join P4 identities to retained V6.7 multi-horizon outcome rows through the official nuScenes scene order and DriveStudio instance tokens. Only 5 P4-train Actors from two scenes align, so we reject joint fitting rather than recycle calibration/test Actors. The frozen descriptive audit instead contains 88 test Actors and 39,285 rows (Table 4). P4 selects 5 Actors and abstains on all 23 whose compiled Chamfer is worse than the query; selected-and-harmful is therefore zero. Its 1,781 selected rows contain no false-safe or decision-flip events. Nevertheless, selected q90 motion error is 1.577 m versus 0.693 m for abstained Actors (2.28×), reaching 4.246 m at 3 s. The physical gate protects representation quality but cannot replace the downstream reliability authority. These are retained source outcomes, not a causal effect of repair or an execution of the reliability model.

**Qualitative protocol.** Main-paper and supplement scenes are frozen from metadata and registered failure strata before rendering method outputs. Panels show RGB, LiDAR, physical surface, ray evidence, compiler action, and the repaired world. We report all frozen cases, including failures, rather than selecting only visually favorable examples.

# 6. Limitations

HARP-3D is not a formal guarantee of real-road safety. Sparse or one-sided sensing can leave large regions unknown, and deterministic completion is intentionally conservative. Actor cuboids and tracked identities may themselves be wrong. The selective model has only 29 admissible nuScenes training Actors; its calibration coverage changes from 8.9% on nuScenes to 75.6% on AV2, directly precluding a cross-domain exchangeability claim. Its population false-repair bound is not conditional risk among selected Actors. Attribution identifies a concrete sensor-opportunity shortcut: observation-frame count accounts for 49.5% of AV2 validity sensitivity. This is model sensitivity rather than causality, but it prevents a domain-invariance claim and motivates opportunity-normalized features evaluated on a fresh external cohort. Exact identity alignment supplies only two P4-train scenes and five Actors, so we do not jointly fit the physical and reliability heads. Zero false-safe events among five selected test Actors is descriptive, not a confidence bound. Interface, representation, dynamics, and calibration shift otherwise remain unresolved. The current physical compiler targets Actor-local surfaces; static topology and deformable Actors remain incomplete. Finally, preservation of identity, trajectory, and size guarantees only hazard functions that depend on those variables; appearance-dependent human judgments may still change.

# 7. Conclusion

We formulate driving-world compilation around a simple boundary: validity is not difficulty. HARP-3D repairs evidence-inconsistent physical geometry while preserving legitimate hazardous Actors and factorizes repairability from hazard with explicit abstention. Frozen nuScenes-to-AV2 transfer reduces false repair without using hazard inputs or deleting hazardous Actors. Its audits turn two abstract guarantee gaps into measurable boundaries: cross-domain validity relies strongly on sensor opportunity, and physically admissible repairs concentrate on Actors with wider long-horizon motion error. SceneIR therefore keeps the physical-repair gate separate from multi-horizon task authority, alongside appearance, collision state, unknown space, and provenance. Remaining work targets opportunity-robust transfer and fresh-domain validation of both authorities without expanding the demonstrated claim boundary.

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